

Soham Deshpande

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Objective

AI & ML postgraduate student with experience developing intelligent systems in computer vision, natural language processing, and remote sensing using Python and modern AI frameworks. Seeking entry-level AI/ML or Data Science role to apply analytical thinking, software engineering principles, and machine learning expertise to solve real-world problems.

Projects

1. InfraNova AI — Thermal Infrared to RGB Satellite Image Translation(Currently Working)

- Trained a Pix2Pix cGAN (Dynamic U-Net + PatchGAN) to colorize Landsat 9 thermal infrared satellite imagery into RGB, boosting downstream YOLO detection confidence on nighttime data.
- Built an end-to-end data pipeline with Google Earth Engine ingestion, anomaly filtering, and region-level stratification across ~10K paired patches.
- Deployed an interactive Streamlit demo with test-time augmentation, Docker support, and Kaggle-ready reproducible training.

2. Translytic — AI-Based Real-Time Video Captioning & Translation System

- Developed a real-time video captioning and translation system using Python, OpenAI Whisper, Gemini API, Google Translate API, edge-tts, OpenCV, and Tkinter.
- Built an end-to-end AI pipeline for speech-to-text, translation, subtitle generation, and multilingual dubbing with timestamp synchronization.
- Integrated REST APIs and JSON-based data processing with automatic fallback mechanisms for transcription and translation.
- Implemented multithreaded video playback, frame buffering, and audio-video synchronization for smooth real-time performance.

3. NoteFlow — Full Stack Notes Management Platform

- Developed a full-stack notes management application using Django, MySQL, HTML, CSS, and JavaScript.
- Implemented user authentication, CRUD operations, category-based filtering, and role-based access control.
- Designed relational database schemas using Django ORM and MySQL with support for per-user notes and dashboard analytics.
- Built AJAX-based pin functionality and real-time search capabilities without full page refreshes.

Education

Master of Computer Applications (AI & ML)
Ramdeobaba University, Nagpur | 2025 – Present

Bachelor of Computer Applications
G.H. Raisoni College of Engineering and Management, Nagpur | 2022 – 2025

12th Grade (HSC)
Pratibha Junior College (Maharashtra State Board) | 2022

10th Grade (SSC)
S.N.B.P. International School (CBSE) | 2020

Skills

Technical Skills

- Programming Languages: Python, SQL
- AI/ML Frameworks & Libraries: NumPy, Pandas, OpenCV, Matplotlib, PyTorch, Scikit-learn
- Generative AI & APIs: OpenAI Whisper, JSON, edge-tts
- Web Development: Django, HTML, CSS
- Databases: MySQL, Oracle Apex
- Developer Tools: Git, GitHub, VS Code, Google Colab, Streamlit
- Core Concepts: Data Structures, Algorithms, Object-Oriented Programming, Database Management Systems

Soft Skills:

- Analytical Thinking
- Problem Solving
- Communication
- Team Collaboration

Certifications

- Google Cloud Career Launchpad – Computing Foundations Track, Google Cloud (Apr 2026)
- Assets, Threats, and Vulnerabilities – Google Cybersecurity Certificate, Coursera (Apr 2026)
- Connect and Protect: Networks and Network Security – Google Cybersecurity Certificate, Coursera (Mar 2026)
- Tools of the Trade: Linux and SQL – Google Cybersecurity Certificate, Coursera (Apr 2026)